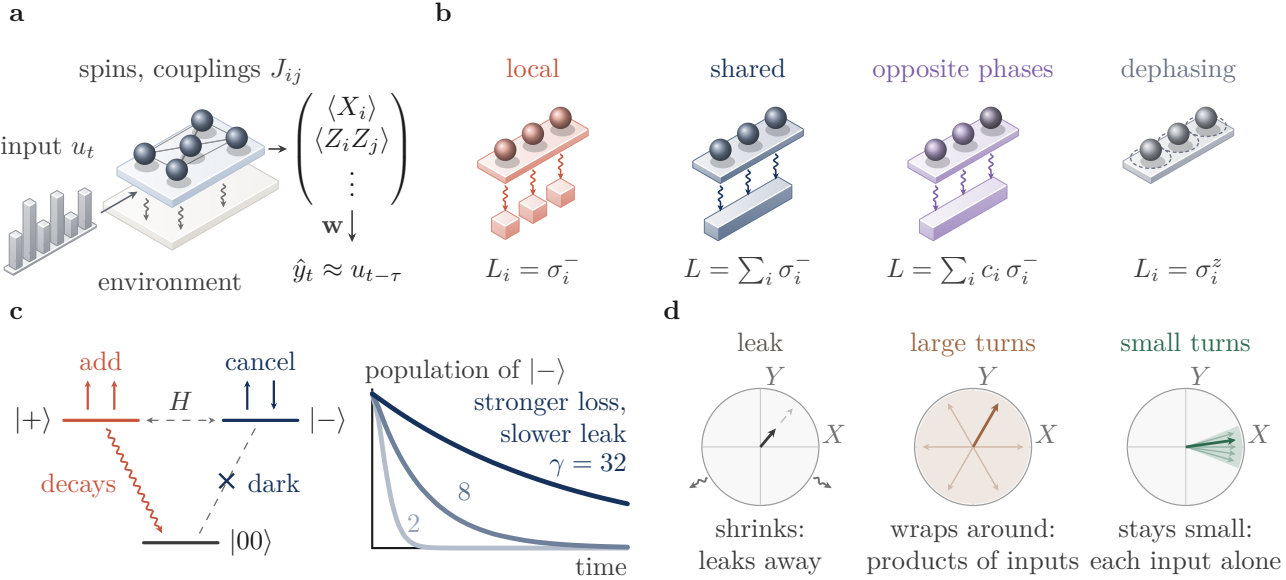

a
spins, couplings J_{ij}
input u_t
environment
$\langle X_i \rangle$
$\langle Z_i Z_j \rangle$
$\mathbf{w}$
$\hat{y}_t \approx u_{t-\tau}$
b
local
shared
opposite phases
dephasing
$L_i = \sigma_i^-$
$L = \sum_i \sigma_i^-$
$L = \sum_i c_i \, \sigma_i^-$
$L_i = \sigma_i^z$
c
add
cancel
H
$|+\rangle$
$|-\rangle$
decays
dark
$|00\rangle$
population of $|-\rangle$
stronger loss,
slower leak
$\gamma = 32$
8
2
time
d
leak
large turns
small turns
Y
X
shrinks:
leaks away
wraps around:
products of inputs
stays small:
each input alone